WEB PRODUCTS · AI SYSTEMS · DEVELOPER PLATFORMS

Carlos Guerrero

Software Engineer | Web Products, AI & Platforms

Asunción, Paraguay hi@carlosguerrero.com +595 985690905 github.com/guerrerocarlos

PROFILE

I build products and reusable systems that help teams deliver. My work spans React and TypeScript interfaces, real-time AI experiences, and deployment platforms that make infrastructure easier to use. I take ownership across product, backend, data, and infrastructure teams, adapting to what the company needs as products evolve.

EXPERIENCE

rPotential

March 2025 - August 2026

Software Engineer, AI Products & Platform

Joined early to build AI-powered decision-support products for executives, then expanded into shared infrastructure and integration across frontend, backend, and data teams.

- Built reusable frontend and backend templates adopted by three teams: frontend, backend, and data. Enabled teams to launch projects without managing the underlying deployment infrastructure.
- Built most initial prototypes in React and TypeScript, using Zustand for state management. Approximately half evolved into products before further iteration.
- Created real-time AI voice experiences that generated data presentations in response to user requests, addressing latency challenges to improve conversational responsiveness.
- Developed Slack-integrated AI for data gathering and user interaction, plus a Twilio and OpenAI Realtime telephone prototype to collect employee information and feedback.
- Connected frontend applications to evolving backends and data pipelines, collaborating on integrations requiring restricted source-data access and anonymized downstream information.

Anaconda

March 2024 - December 2025

Applied AI Engineer

Built serverless infrastructure for AI agents and LLM workloads, supporting deployment, orchestration, and integration through OpenAI-compatible interfaces.

- Created reusable deployment workflows and queue-based communication to support rapid agent iteration and asynchronous multi-agent workflows.
- Delivered a Slack-integrated agent connected to Jupyter, helping scientists improve notebooks and execute Python from Slack.
- Added tracing and monitoring with AWS X-Ray and CloudWatch to support performance analysis and debugging.

Masterworks

January 2018 - January 2020

Backend Engineer to DevOps Lead

Moved from backend engineering into DevOps leadership, building shared infrastructure to improve developer experience and deployment reliability.

- Created automated per-branch environments so teams could develop and validate changes independently.
- Migrated frontend hosting to AWS CloudFront, supporting CDN-level A/B testing and performance tuning.
- Built and maintained frontend-related microservices for serverless infrastructure, including a progress bar still used in marketing emails.
- Reworked backend delivery from Jenkins and Kubernetes to GitHub Actions and AWS Fargate, with infrastructure managed through Terraform and AWS CDK.

SELECTED PLATFORMS & PROJECTS

InglésConLiza

Creator & Developer | Featured personal project www.inglesconliza.com

Built an integrated language-learning platform for my wife, connecting the learner experience, customer communications, content operations, and media delivery through a microservices architecture.

- Created the original Inbox implementation, bringing Instagram, Messenger, WhatsApp, Telegram, email, and in-app chat into a shared customer communications workflow.
- Developed supporting authentication, accounts, mailer, and email-template services, plus a content manager integrated with Google Drive.
- Built video processing with Cloudflare Containers and FFmpeg.

W7S.io

Creator & Developer w7s.io

Built an open-source platform for deploying full-stack applications from GitHub. Unified hosting, backends, databases, queues, and workflows with OIDC authentication, usage controls, and infrastructure inspection for developers and AI agents. Built dashboard interfaces with React Router and Framer Motion.

Inbox

Creator & Developer omattic.com

Created the original React and TypeScript communications platform for InglésConLiza, unifying channels, real-time conversations, and AI-assisted workflows. Used TanStack Query for conversation caching and prefetching, with TanStack Router for navigation. Campaign integration is planned.

Invoices Templates

Creator & Developer www.invoices-templates.com

Built a complete React and TypeScript invoice editor with Radix UI/shadcn components and Tailwind CSS end to end, combining rich-text customization, local drafts, authenticated saving, and consistent print/PDF output in one document workflow.

SEOKeywordExplorer

Developer www.seokeywordexplorer.com

Developed a React and TypeScript keyword research application combining DataForSEO and Google Trends data, keyword reports, and historical search positions. Cached reports avoid repeated provider requests and credit charges unless explicitly refreshed.

Democrachain

Creator & Developer | Independent platform and working prototype github.com/democrachain

Designed and built a JavaScript/TypeScript blockchain prototype combining shared data synchronization, scoped permissions, and smart contracts to reduce coordination work for distributed application teams.

SMALL PERSONAL PROJECTS

Compress Video

Built a browser-based video tool for resizing, trimming, cropping, and joining videos, with processing performed locally and no backend required.

KeePassWeb

Built a frontend password-encryption tool using browser-native cryptography, allowing users to encrypt and decrypt credentials locally without a third-party password-storage service.

Secure Channel

Built a backend-free credential-sharing tool for people and AI agents using Web Crypto and browser-local private keys. Added a dependency-free offline Node.js sender, agent-readable instructions, and copy-friendly recipient links that preserve compatibility with existing keys.

OLDER PROJECTS

- **CacheP2P**: Browser-based JavaScript caching library using WebTorrent to share website content between visitors and reduce origin-server requests.
- **TorrenTV**: Desktop torrent-streaming application built with Node.js and node-webkit, streaming video to Apple TV via AirPlay.
- **JS Bitcoin Miner**: Educational JavaScript/Node.js miner demonstrating block hashing, difficulty targets, and nonce verification for Bitcoin and Bitcoin Cash.

TECHNOLOGIES & ENGINEERING

React · TypeScript · JavaScript · Zustand · TanStack Query · TanStack Router · React Router · Radix UI / shadcn/ui · Tailwind CSS · Framer Motion · Vite · Node.js · Python · REST APIs · Real-time AI · GitHub Actions · AWS · Cloudflare Workers · CI/CD · Distributed systems

EDUCATION

Universidad del Táchira

2002 - 2007

Electronic Engineering

Coursework: C++ · Process Automation · Digital Signal Processing · Control Systems · Microcontrollers · Electronics

ETI

1999 - 2002

Industrial Chemistry

Coursework: Chemical Engineering · Organic Chemistry · Inorganic Chemistry · Analytical Chemistry · Physical Chemistry · Biochemistry

TRAINING & CERTIFICATIONS

AWS Certified Cloud Practitioner

Amazon Web Services Training and Certification | 2020

Cisco Certified Network Associate (CCNA)

CISCO - Networking Academy | 2015

TEACHING & INSTRUCTION

- Delivered MongoDB courses as an instructor.
- Delivered Blender courses as an instructor.

LANGUAGES

English: fluent | Spanish: native